

CE 310 — Capstone Report

Team submission | Due: Saturday, December 6 by midnight — submit to D2L

Overview

Your team's in-class exercise notebook is where you ran the analysis. This report is different: it's where your team steps back from the code and writes the deliverable your client actually asked for — a short, professional document that answers their specific questions in plain language, backed by your numbers.

Before writing this report, **make sure all four of you can explain the numbers in your notebook** — the Model 2 coefficients, the t-test output, the Monte Carlo results. Every member presents a portion in Week 16 and takes questions on it, so a number nobody at the table can account for is a number to revisit now.

How to Submit

Read this before you submit. Following this submission format exactly is essential — with a class this size, grading depends on it. Submissions that don't comply will be penalized.

Requirement	Detail
Format	PDF, one file per team
Length	2–3 pages (excluding an optional appendix)
Submission	One team member submits on behalf of the whole team. List all four full names at the top of the report.
File name	lastname1_lastname2_lastname3_lastname4_capstone_report.pdf (alphabetical by last name)
Deadline	Sun Dec 6, 11:59 pm — D2L Dropbox

Required Content

Your own client brief (your client brief) lists the 2–3 specific questions your team wrote for your stakeholder. Your report must answer **every one of them explicitly** — not just present your analysis and hope the client can connect the dots.

- 1. Executive Summary (3–5 sentences)** The bottom line, up front: what did you find, and what should the client do? A busy stakeholder should be able to read only this paragraph and walk away with the key takeaway.
- 2. Dataset & Client Overview (short paragraph)** What the dataset is and where it's from, who the client is, dataset size (ANSWER_A1 rows), and the mean of your outcome variable (ANSWER_A3) in plain terms.
- 3. Findings — answer each client question directly** For each numbered question in your client brief, write a short subsection: state the question, then answer it using your team's regression (Model 1 vs. Model 2), hypothesis test, and uncertainty/Monte Carlo results. Include the specific numbers (coefficients, p-value, R^2 , P10/P90 range) — a client-facing report still needs to show its work, just without code or statistical jargon left unexplained.
- 4. Recommendation** One concrete, quantified, actionable recommendation that follows from your findings (e.g., a specific operational change and its estimated savings, or a specific risk to budget for given the P10/P90 range). Avoid generic advice (“reduce energy use”) — tie it to a number from your analysis.

5. Appendix (optional, doesn't count toward the 2–3 page limit) A table of key statistics (Model 1/Model 2 coefficients, R^2 , t-statistic, p-value, cost P10/P50/P90) for reference, if you want to keep the main body number-light.

Using AI Tools

The course policy applies here as everywhere else (see the syllabus): use AI to learn, attribute it when it wrote what you hand in, and own what you submit either way. Two things are specific to this report.

Attribution is per team. The syllabus asks for a brief note naming the tool and what it produced. On a team document, put one note at the end of the report covering the whole team's use — you do not each write your own.

Every number must be in your notebook. The report cites figures your analysis produced; a preceptor will pick two and look for them in your notebook output. This is the check that matters, and it is the reason a model cannot write this report for you: it can produce fluent paragraphs about regression, but it cannot produce *your* coefficient on *your* data, and a report whose numbers do not appear in your notebook is a report you cannot defend in Week 16.

What This Is Not

- Not a copy-paste of your notebook's Section E memo — that was brief (3 paragraphs) and written for the notebook; this is a fuller, client-facing document.
 - Not a place for new analysis — use the numbers your team already produced in the in-class exercise notebook.
 - Not graded on length — a tight, well-organized 2 pages beats a padded 3.
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Grading

See the capstone report rubric. Worth 20 raw points, entered in the **Capstone Project + Presentation** category (15% of the course grade) alongside your team's in-class exercise notebook (30 pts) and your team's presentation (50 pts, Week 16) — 100 points in total.

The report is 20 points rather than 30 because ten of them moved to the presentation's Q&A. Length was never the point — a tight two pages that answers your client's questions with your own numbers scores everything available here.